

TsaoSciResearcher Engineering Audit

Release 0.5.3 | current-tree evidence | 2026-07-23

Evidence-first scientific research control layer

Question -> evidence -> design -> execution handoff -> validation -> artifact

Truth boundary: a contract or handoff is not an execution receipt.

340

Capability contracts

322

Named catalog contracts

15

Workflows

15

Schemas

25

Test modules

7

Domain packs

Delivered integration

1. Measurement Boundary with calibration, uncertainty and applicability fields
2. Structure-Property planning with confounders, alternatives and scale bridges
3. Bilingual causality guard and evidence-to-source execution-receipt checks
4. Deterministic test/quality HTML-SVG interfaces and this four-page PDF

Repository operation

All changes were written directly to main; no branch and no pull request were created.

Scoped software validation matrix

Recorded cross-platform baseline

These platform results are the last complete baseline, not a fresh current-tree run.

macos python 3 12		PASS
ubuntu python 3 10		PASS
ubuntu python 3 13		PASS
windows python 3 12		PASS

Baseline and focused current-change gates

bandit high severity		PASS
bounded performance		PASS
byte identical release builds		PASS
complete regression		PASS
critical mutation killed		15/15
deterministic visual reports		PASS
json schemas 15		PASS
mypy strict		PASS
repository and contract audit		PASS
reverse order regression		PASS
ruff format and lint		PASS
scientific quality guards		PASS
seeded random order regression		PASS
validation tree digest		PASS

Composite PASS means the declared scopes passed; current end-to-end CI remains NOT RUN.

Scientific quality controls

Four deterministic contracts: measurement, multiscale reasoning, causal claims and evidence provenance.

Examples: PASS 3 | WARN 0 | BLOCK 1 | average completeness 89%

Measurement boundary **PASS**

Declared completeness: 100%



Operational definition, calibration, uncertainty and applicability are explicit.

MB-COMPLETE: Operational measurement boundary is explicit.

Kind: measurement-boundary

Structure-property planner PASS

Declared completeness: 100%



The chain records intervention, structure, mediator, response and competing...

SP-CHAIN: Structure, mediator and property are all explicit.

Kind: structure-property-plan

Causality guard **BLOCK**

Declared completeness: 57%



Association-only evidence blocks causal wording in English and Chinese.

CG-OVERCLAIM: Causal wording exceeds the declared study design.

Kind: causality-guard

Evidence traceability	PASS
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Declared completeness: 100%



Claims link to evidence locators and execution claims require receipts.

ET-COMplete: Claim, evidence, locators and execution boundary are traceable.

Kind: evidence-traceability

BLOCK is intentional when wording exceeds evidence or an execution claim lacks a receipt.

The guards remain configurable and do not hard-code material-specific trends.

Closed-loop evidence and limitations

Recorded full-repository baseline

GitHub Actions run: not recorded

All engineering/scientific gates passed before publication transport failed.

Focused current-change regression

Tests: 0 passed / 0 failed | environment not recorded

Scope: scientific quality, report generators and visual data contract.

Current-tree end-to-end CI

Status: not recorded

No current source-tree digest or fresh cross-platform receipt is asserted.

Explicit limitations

Reproduction entry points

```
python -m pytest -q tests/test_scientific_quality.py  
tests/test_visual_report_contract.py
```

```
python scripts/build_test_dashboard.py --check
```

```
python scripts/build_research_quality_dashboard.py --check
```

```
python scripts/build_engineering_report.py --check
```

Software validation is not acceptance of experiments, external calculations, legal or safety claims.